AV AI VELOCITY ARCHITECT INTELLIGENCE · ISSUE 04

WEEK OF 19–25 SEPTEMBER 2026

# After The Yes

A reviewer approved two writes and one ran zero times. A failing write ran twice. A vendor put in writing that recovery is not exactly-once.

**22** repos read

**3** mechanisms executed

**8** papers read

**17** models priced

**Vijay Zharotia**
Healthcare AI architecture

WHY AI VELOCITY

# Agents that act inherit obligations.

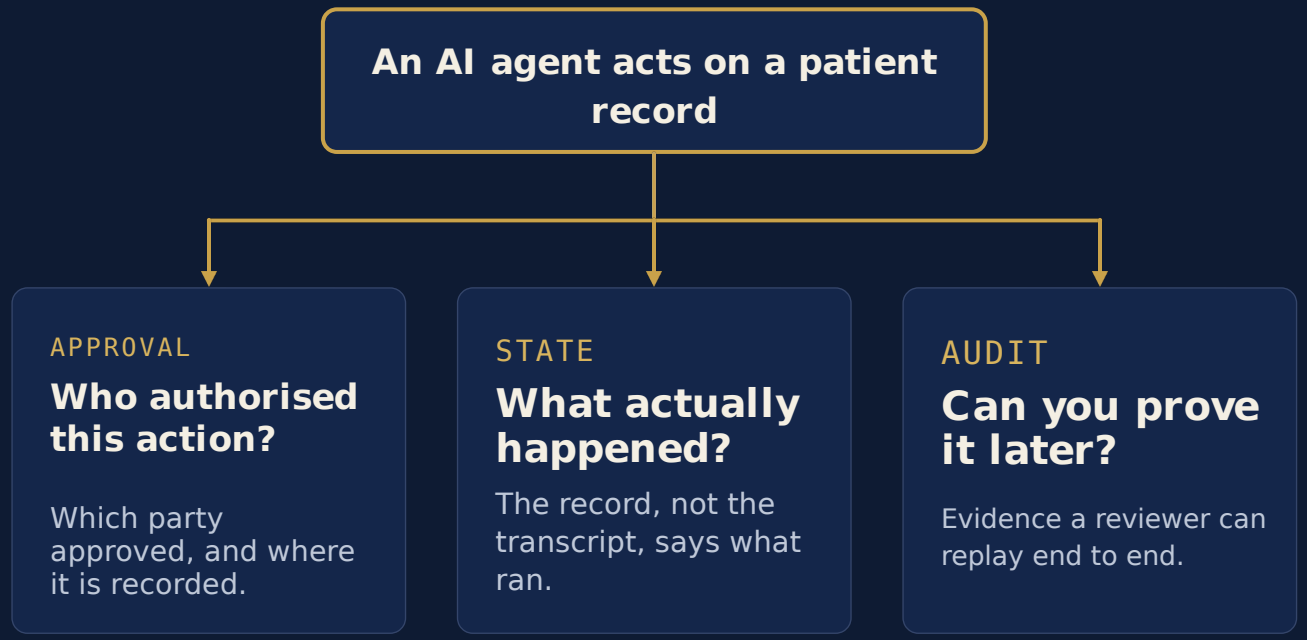
A weekly architecture brief for teams putting AI agents into healthcare workflows. We read the releases, run the mechanism, and say what to change.

FOR ARCHITECTS

The mechanism, the code, the move.

FOR PRODUCT MANAGERS

The decision, the risk, the date.




EVERY CLAIM CARRIES ITS EVIDENCE

**EXECUTED**

We ran it. The numbers are observed output.

**DOCS**

Read from documentation or source.

**UNVERIFIED**

Not confirmed. Vendor numbers carry an asterisk.

THE WEEK IN ONE VIEW

# From what the yes covers to how often it runs

CONTEXT

## Last week: the yes bound to arguments

Issue 03 asked which arguments a clinician’s approval covers. This week asked a later question: how many times the approved write runs.

→

PROBLEM

## Zero, two, or not promised

An approved write ran zero times. A failing write ran twice. A vendor wrote that recovery does not imply exactly-once execution.

→

SOLUTION

## Join approvals to results

Match every approval to a success result by call id before anything is reported as done, and key every write so it cannot land twice.

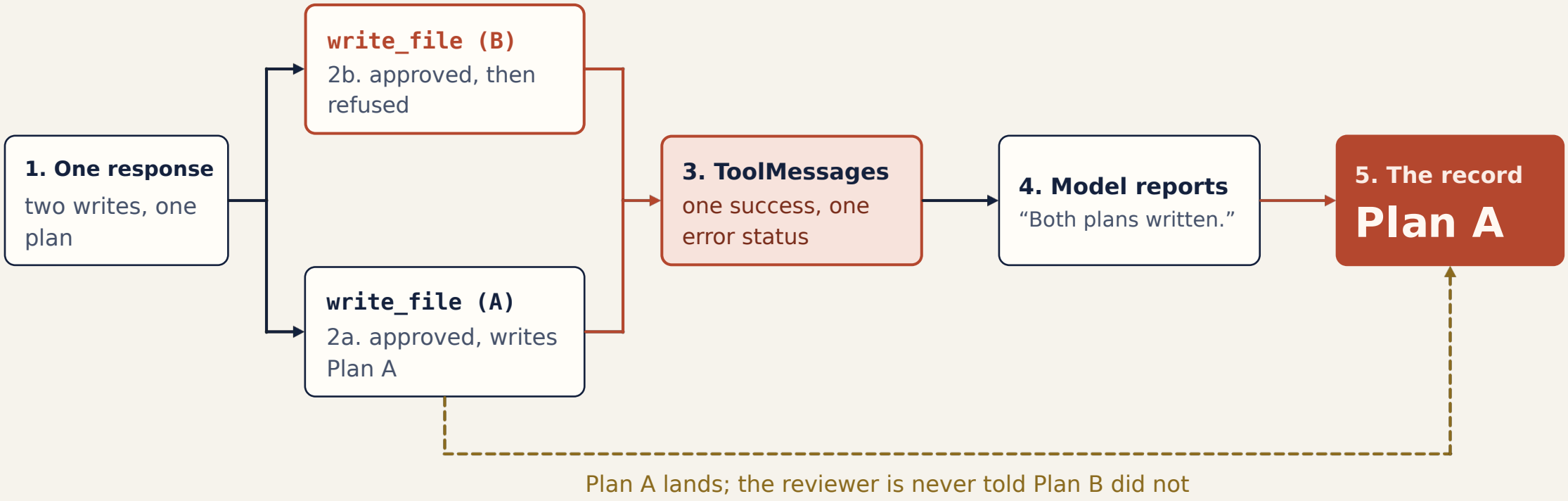
EXECUTION · 04–06
How many times it runs

DATA · 07
Secrets outside the policy

EVIDENCE · 08
Rows nobody counted

COST, DATES · 09–10
Repricing and dockets

# The approved write that never ran



EXECUTED MECHANISM · SOLUTION · YOUR HARNESS

# Join every approval to its result

STEP 1
**Resume after approval**
Collect every approved call id.

→

STEP 2
**Join to results**
Match each to a success ToolMessage by call id.

→

STEP 3
**A miss is an incident**
Not a retry. Tell the reviewer.

→

STEP 4
**Re-run plan evals**
First-wins is a different system from last-wins.

**2 to 1**
writes executed for 2 approvals, 0.7.15 to 0.7.16

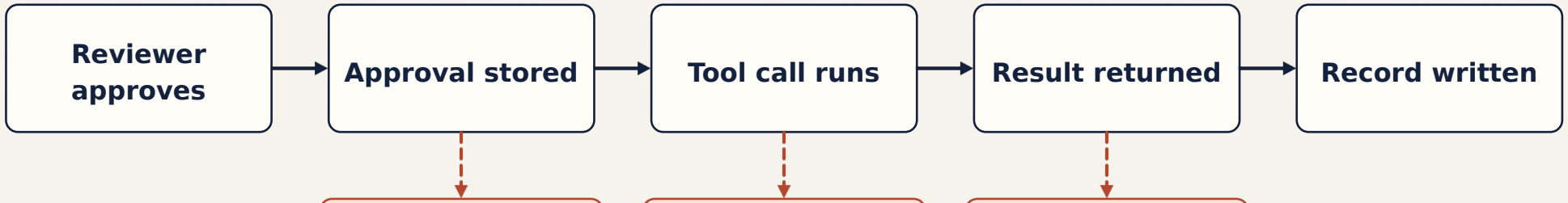
**0**
notices to the reviewer that an approved call was refused

**1st wins**
the parallel write that wins the record; it used to be the last

> test_da_parallel_writes.py runs both versions in under two minutes. The model said “Both plans written.” on both.

# Three places the count went wrong

**CONTEXT**
how it runs




**PROBLEM**
where it broke

**LANGGRAPH 1.2.12**
A dict schema let ‘aprove’ pass as a no
EXECUTED

**LLAMA-INDEX**
Below 0.14.25 a failing write ran twice
EXECUTED

**MS AGENT FW**
A child’s pending approval now fails the parent
DOCS

**SOLUTION**
what to build

**Key every write, type every approval**

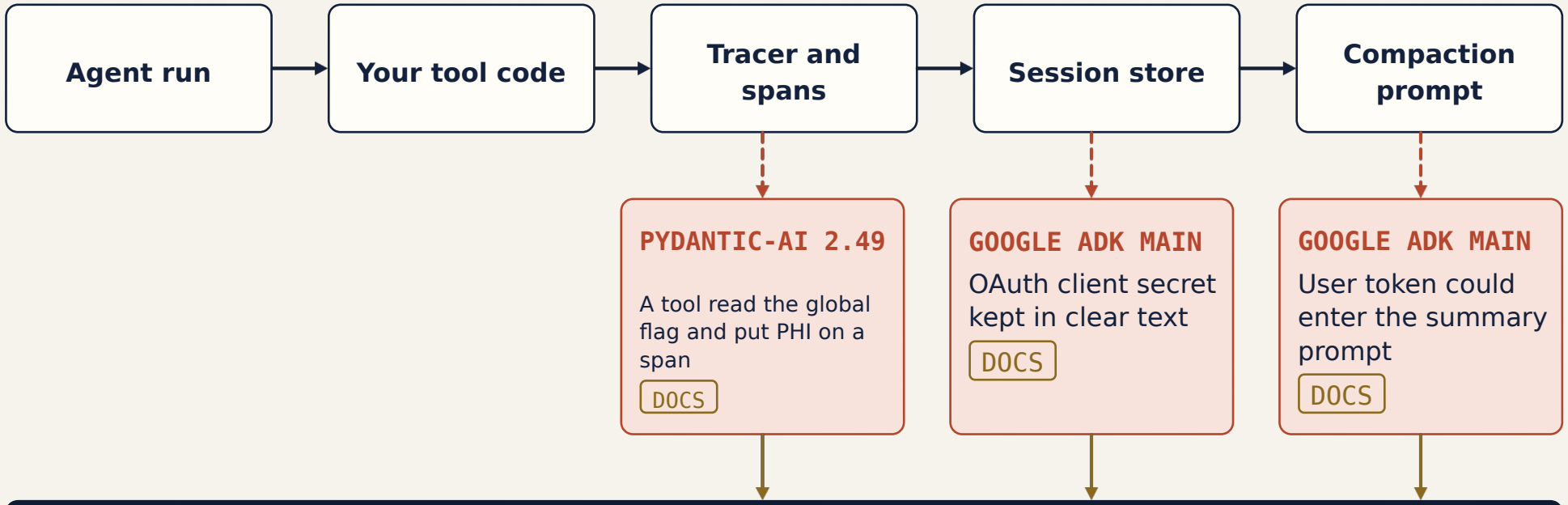
Give every write an idempotency key the tool checks before it commits. Validate approvals with a Pydantic model, never a dict, and make the key cover retries.

# Secrets and PHI found a second path

**CONTEXT**
how it runs

**PROBLEM**
where it leaked

**SOLUTION**
what to build

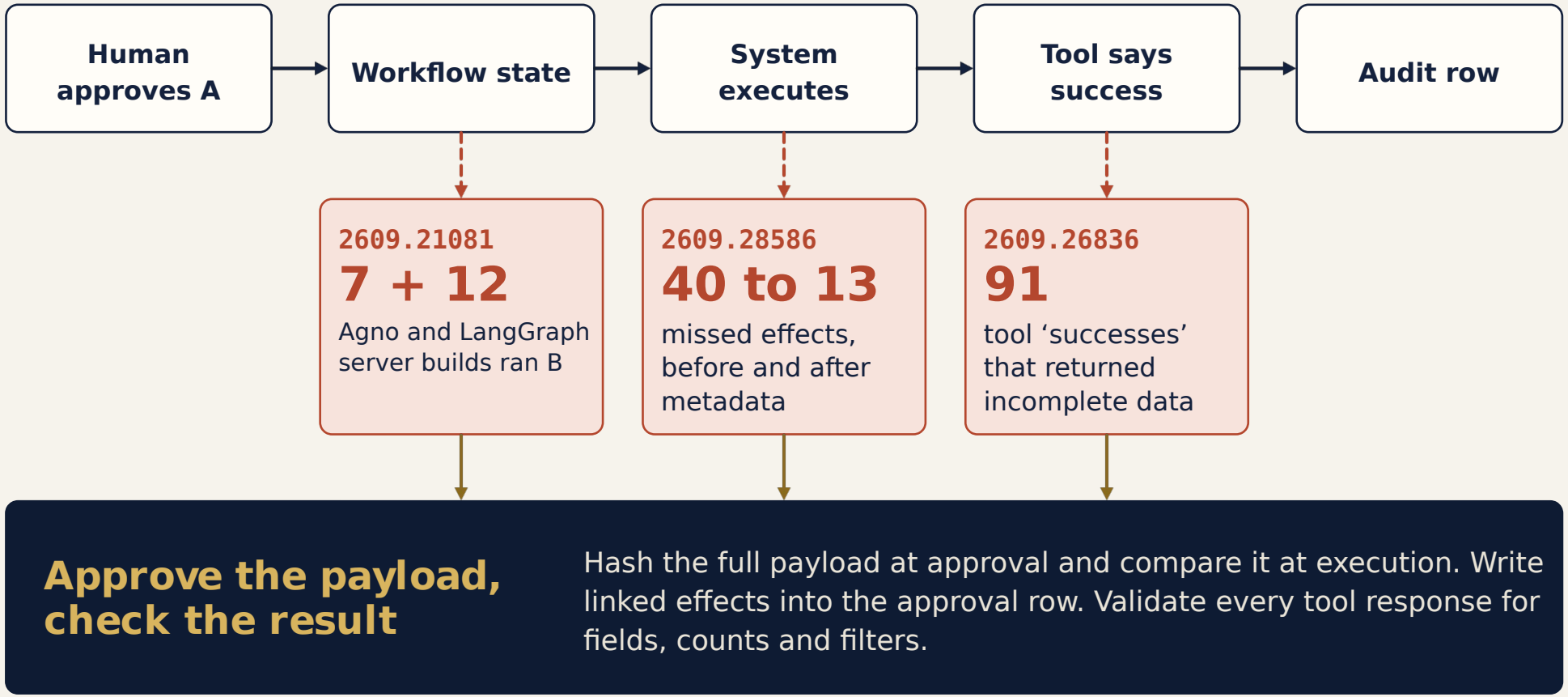


# Approve A, execute B, count nothing

**CONTEXT**
how it runs

**PROBLEM**
where it broke

**SOLUTION**
what to build



COST · LIST PRICES READ 25 SEPTEMBER

# Opus 5.5 leads both indexes at $260

| MODEL | COST PER 10,000 DOCS | GEN / MED* |
|---|---|---|
| **GPT-6 Luna** | **$6.50** | not read |
| **GPT-6 Sol** | **$130** | **48 / 43.5** |
| **GPT-5.6 Sol** | **$260 deprecated** | **47 / 45.4** |
| **Claude Opus 5.5** | **$260** | **58 / 60.5** |
| **Claude Opus 5** | **$325 deprecated** | not read |

Per million tokens in/out: Luna $0.10/$0.50, Sol $2/$10, GPT-5.6 Sol $4/$20 (promotional), Opus 5.5 $4/$20, Opus 5 $5/$25. 4,000 in and 500 out per document, uncached. *Artificial Analysis general and Healthcare & Medical scores as read 25 Sep, not re-verified; Opus 5.5 now reads 61 on medical.

**CONTEXT**

Opus 5.5 and GPT-6 Sol and Luna all launched on 22 September.

**PROBLEM**

Sol beat its predecessor on the general index and fell on the medical one. Claude’s newer tokenizer makes about 30% more tokens for the same text.

**DECISION**

Re-tokenise your own documents, price the cached run, and record both scores with the date read.

# Two FDA comment windows and a workshop




● REGULATORY ● MARKET AND FUNDING

*Date not confirmed; awards reported 30 Sep to 1 Oct.

# Signed evidence found a health-sector home

CHAI

### Stewardship of OVERT

An open spec for signed runtime evidence of agent tool calls, taken over from Glacis.

LANGCHAIN

### Release gates in healthcare

Abridge’s five-stage gates and clinician annotation queues; Trajectories in LangSmith.

MICROSOFT

### Recovery is not exactly-once

Checkpoint-and-recover hosting, plus Foundry egress policies in Audit and Enforced modes.

UK AISI

### Verified transcripts

Five benchmarks including HealthBench across six frontier models, in a shared schema.

OPENAI

### Rules for outside assessments

Seven principles, and MentalHealthBench with 80-plus clinicians across 19 languages.

QUIET THIS WEEK

HL7, FDA on generative AI, ONC, CMS, NIST, Epic, MCP. Oracle Health launched write-side agents with no published audit contract.

# Seven moves, with the reason for each

| Move | Why | By |
|---|---|---|
| Join each approved call to a success result by call id | Refused writes go unseen | Every resume |
| Add an idempotency key every write checks before commit | A failing write ran twice | This sprint |
| Validate approvals with a Pydantic model, never a dict | A typo became a no | Next upgrade |
| Rotate OAuth client secrets on durable session stores | Stored in clear text | Now |
| Hash the full payload at approval; compare at execution | Approve A, execute B | Design review |
| Re-tokenise your own documents before budgeting Opus 5.5 | About 30% more tokens | Before budget |
| File on the FDA robotics and gen-AI dockets | Autonomy vocabulary set here | 19 Oct, 24 Nov |

AV AI VELOCITY

THIS WEEK'S POSITION

# An approval is not a write until you join it to the result.

**Long-form edition**
Every mechanism, source and date

**Lab package**
Runs clean, no API key needed

**New issue every Friday**
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github.com/zkumar/ai-velocity-weekly